Mehrazin Alizadeh

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Summary

Applied Machine Learning Scientist with 5+ years of experience across ML research, production systems, and real-world decisioning. Experienced in causal/uplift modeling, treatment-effect estimation, recommendation/ranking, and constrained optimization over large-scale banking data, alongside end-to-end LLM systems spanning RAG, tool-calling agents, evaluation frameworks, post-training, and efficient inference. First-author IEEE publications in differentiable projection and constraint-aware deep learning, with research contributions to LLM compression.

Experience

Applied Machine Learning Scientist II Jul 2025 – Present
TD Bank AI2 *Toronto, Canada*

- Core technical owner of **NBE (Next Best Engagement)**, a large-scale offer-ranking system scoring 16M customers across 37 products using PRISM/legacy propensities, calibration, CLTV, and campaign-response signals.
- Led **multi-touch attribution and personalized channel-decisioning** research from problem formulation through customer-journey analysis, modeling, and evaluation; investigated treatment-selection bias and potential confounding in historical campaign data.
- Built a **cross-fitted doubly robust (DR-learner) uplift model** with per-channel propensity models to estimate conditional treatment effects for personalized channel selection; currently developing a **transformer-based causal/uplift approach over longitudinal customer journeys** to model sequential treatment-response dynamics.
- Conducted **PRISM impact analysis** for senior stakeholders, assessing NBE's migration from legacy to foundation-model propensity scores across ranking shifts, coverage, and calibration stability.
- Extended ORCA, the team's in-house optimization framework, for scalable product decisioning under acquisition, NPV, eligibility, capacity, and business-mix constraints.

Machine Learning Engineer Sept 2023 – Jul 2025
Multiverse Computing *Toronto, Canada*

- Served as technical lead across multiple GenAI initiatives, including an agentic medical research QA system with tool-calling workflows for paper retrieval, query rewriting, memory, hybrid search, response synthesis, and adaptive execution.
- Designed a reusable LLM evaluation and synthetic-data framework for RAG systems, improving internal answer-quality evaluations by roughly 20 percentage points across retrieval, reranking, grounding, generation quality, and latency diagnostics.
- Built distributed LLM pretraining and post-training infrastructure supporting SFT, DPO, LoRA, QLoRA, ZeRO-3 training, checkpointing, and MLflow tracking; validated multi-node runs for models up to 70B scale.
- Contributed to **CompactifAI**, Multiverse's in-house LLM compression technique, validating compressed models for domain QA, deployment readiness, and model-efficiency research.
- Owned end-to-end delivery of a fully local multilingual RAG chatbot for technical support, integrating hybrid retrieval, reranking, session-aware query rewriting, vLLM inference, offline deployment, and internal compressed-model validation.
- Led an on-device automotive voice assistant from idea conception to functional prototype, combining real-time STT/RAG/TTS, personalized prompting, compact LLM inference, asynchronous streaming, and judge-based benchmarking for small-model deployment feasibility.

Machine Learning Research Assistant Jan 2021 – Aug 2023
York University *Toronto, Canada*

- Researched **constraint-aware deep learning** for QoS-guaranteed wireless resource allocation, designing differentiable projection methods that enforce feasibility while preserving gradient-based training.
- Proposed **DIPNet** and **DEPNet**, projection-based neural architectures for constrained power control, achieving zero QoS constraint violation and orders-of-magnitude faster inference than classical optimization baselines.
- Developed **GNN-based resource allocation models** for IRS/STAR-RIS networks, leveraging graph structure to improve scalability across variable-size wireless network topologies.
- Built reproducible ML research pipelines in PyTorch, TensorFlow, Docker, and W&B for simulation, model training, ablation studies, and paper-ready experimental validation.
- Guided junior researchers on projection-based deep learning experiments and authored first-author IEEE TCOM and IEEE Wireless Communications Letters papers on constraint-aware ML for wireless optimization.

Education

Master of Applied Science in Computer Engineering Jan 2021 – Nov 2022
York University *GPA: 3.92 / 4.00*

Bachelor of Science in Electrical Engineering, Telecommunications Sept 2015 – Aug 2020
University of Tehran *GPA: 3.62 / 4.00*

Technical Skills

Languages & Data: Python, SQL, PySpark, Bash, Pandas, NumPy, Databricks

Machine Learning & Deep Learning: PyTorch, TensorFlow, JAX, scikit-learn, XGBoost, Hugging Face Transformers, calibration, SHAP, experiment design

LLMs, Agents & Evaluation: RAG, tool-calling agents, agentic workflows, LLM-as-judge evaluation, synthetic QA generation, retrieval evaluation, reranker evaluation, grounding evaluation, prompt evaluation, vLLM, LlamaIndex, LangChain, LiteLLM, Ollama

Post-Training & Distributed ML: continued pretraining, SFT, DPO, preference optimization, LoRA/QLoRA, TRL/PEFT, Ray, DeepSpeed, ZeRO-3, MLflow

Causal ML, Decisioning & Optimization: treatment-effect estimation, causal/uplift modeling, doubly robust estimation, constrained optimization, OR-Tools, recommendation/ranking systems, targeting, customer/product decisioning

Cloud & Deployment: AWS, GCP, Docker, FastAPI, Streamlit, Git/GitLab

Selected Publications

- **Mehrazin Alizadeh**, Hina Tabassum, "Power Control with QoS Guarantees: A Differentiable Projection-Based Unsupervised Learning Framework", *IEEE Transactions on Communications*, *2023* (link).
- **Mehrazin Alizadeh**, Xavier Mootoo, Omer Waqar, Hina Tabassum, "QoS-Aware Deep Unsupervised Learning for STAR-RIS Assisted Networks: A Novel Differentiable Projection Framework", *IEEE Wireless Communications Letters*, *2024* (link).
- Andrei Tomut, Saeed S. Jahromi, Abhijoy Sarkar, ..., **Mehrazin Alizadeh**, ..., Román Orús, "CompactifAI: Extreme Compression of Large Language Models using Quantum-Inspired Tensor Networks", *ESANN Proceedings*, *2025* (link).
- Mohammad Amin Saeidi, Hina Tabassum, **Mehrazin Alizadeh**, "Molecular Absorption-Aware User Assignment, Spectrum, and Power Allocation in Dense THz Networks With Multi-Connectivity", *IEEE Transactions on Wireless Communications*, *2024* (link).